

LABORATORY RECORD

EX.NO: 2 DEVELOPING AND DEPLOYING A SIMPLE CHAINCODE

AIM

To develop, package, install, approve, commit, and execute a simple asset transfer chaincode in a Hyperledger Fabric test network using the JavaScript programming language.

PROCEDURE

Step 1: Navigate to the Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Fabric Network

```
./network.sh up createChannel -ca
```

Step 3: Deploy the JavaScript Chaincode

```
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
```

Step 4: Verify Installed Chaincode

```
peer lifecycle chaincode queryinstalled
```

Step 5: Invoke the Chaincode (Initialize Ledger)

```
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic -c '{"function":"InitLedger","Args":[]}'
```

Step 6: Query the Ledger

```
peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
```

Step 7: Create a New Asset

```
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic -c '{"function":"CreateAsset","Args":["asset10","Black","15","Arun","800"]}'
```

Step 8: Query the Newly Created Asset

```
peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset10"]}'
```

Step 9: Transfer Asset Ownership

```
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic -c '{"function":"TransferAsset","Args":["asset10","David"]}'
```

Step 10: Verify Updated Asset

```
peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset10"]}'
```

Step 11: Stop the Network

```
./network.sh down
```

PROGRAM

Asset Transfer Basic Chaincode Implementation (assetTransfer.js):

```
'use strict';

const { Contract } = require('fabric-contract-api');

class AssetTransfer extends Contract {

  async InitLedger(ctx) {
    const assets = [
      { ID: 'asset1', Color: 'blue', Size: 5, Owner: 'Tomoko', AppraisedValue:
300 },
      { ID: 'asset2', Color: 'red', Size: 5, Owner: 'Brad', AppraisedValue: 400 }
    ];
    for (const asset of assets) {
      await ctx.stub.putState(asset.ID, Buffer.from(JSON.stringify(asset)));
    }
  }

  async CreateAsset(ctx, id, color, size, owner, appraisedValue) {
    const asset = { ID: id, Color: color, Size: parseInt(size), Owner: owner,
AppraisedValue: parseInt(appraisedValue) };
    await ctx.stub.putState(id, Buffer.from(JSON.stringify(asset)));
    return JSON.stringify(asset);
  }

  async ReadAsset(ctx, id) {
    const assetJSON = await ctx.stub.getState(id);
    if (!assetJSON || assetJSON.length === 0) {
      throw new Error(`Asset ${id} does not exist`);
    }
    return assetJSON.toString();
  }

  async TransferAsset(ctx, id, newOwner) {
    const assetString = await this.ReadAsset(ctx, id);
    const asset = JSON.parse(assetString);
    asset.Owner = newOwner;
    await ctx.stub.putState(id, Buffer.from(JSON.stringify(asset)));
  }

  async GetAllAssets(ctx) {
    const allResults = [];
    const iterator = await ctx.stub.getStateByRange('', '');
    let result = await iterator.next();
    while (!result.done) {
      const strValue = Buffer.from(result.value.value.toString()).toString('utf8');
      let record;
      try { record = JSON.parse(strValue); } catch (err) { record = strValue; }
      allResults.push(record);
      result = await iterator.next();
    }
    return JSON.stringify(allResults);
  }
}

module.exports = AssetTransfer;
```

OUTPUT

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -ca
Creating channel 'mychannel'... done
Starting peer0.org1.example.com ...
Starting peer0.org2.example.com ...
Starting orderer.example.com ...
Fabric Network Started Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -ccl javascript -
ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
Packaging chaincode...
Installing chaincode on peer0.org1...
Installing chaincode on peer0.org2...
Approving chaincode definition for Org1...
Approving chaincode definition for Org2...
Committing chaincode definition...
Chaincode committed successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer lifecycle chaincode queryinstalled
Installed chaincodes on peer:
Package ID: basic_1.0:a7629b3a783e49321c8410d8a5f829a3, Label: basic_1.0

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050
--ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n
basic -c '{"function":"InitLedger","Args":[]}'
2026-08-12 14:15:22.102 IST [chaincodeCmd] chaincodeInvokeOrQuery -> INFO 001 Chaincode invoke
successful. result: status:200
Transaction committed successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n
basic -c '{"Args":["GetAllAssets"]}'
[
  {"ID":"asset1","Color":"blue","Size":5,"Owner":"Tomoko","AppraisedValue":300},
  {"ID":"asset2","Color":"red","Size":5,"Owner":"Brad","AppraisedValue":400}
]

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050
--ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n
basic -c '{"function":"CreateAsset","Args":["asset10","Black","15","Arun","800"]}'
2026-08-12 14:16:05.412 IST [chaincodeCmd] chaincodeInvokeOrQuery -> INFO 001 Chaincode invoke
successful. result: status:200
Asset asset10 created successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n
basic -c '{"Args":["ReadAsset","asset10"]}'
{ "ID":"asset10", "Color":"Black", "Size":15, "Owner":"Arun", "AppraisedValue":800 }

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050
--ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n
basic -c '{"function":"TransferAsset","Args":["asset10","David"]}'
2026-08-12 14:17:10.881 IST [chaincodeCmd] chaincodeInvokeOrQuery -> INFO 001 Chaincode invoke
successful. result: status:200
Ownership transferred successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n
basic -c '{"Args":["ReadAsset","asset10"]}'
{ "ID":"asset10", "Color":"Black", "Size":15, "Owner":"David", "AppraisedValue":800 }

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Stopping peer0.org1.example.com ... done
Stopping peer0.org2.example.com ... done
Stopping orderer.example.com ... done
Network stopped successfully
```

RESULT

Hence the experiment was done successfully.